Freight Rate Prediction

Machine Learning Engineer Assessment – Validation Approach, Results and December Chart

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1. Summary

The task is to predict the posted rate for 12,000 future loads (Nov–Dec 2025) using 48,000 labelled loads from Jan–Oct 2025. Because the target period lies entirely after the training period, I validated the model in time order rather than with a random split. The final model combines a gradient-boosted tree model for the load itself (route, equipment, weight) with a linear component for the market signals and a time trend. Averaged over three time-ordered folds it reaches a mean absolute error of about $45 (1.9% MAPE) on loads with trustworthy labels, the lowest of all models tested.

2. Train / validation split approach

train_test.csv spans 1 Jan – 31 Oct 2025 and validation.csv spans 1 Nov – 31 Dec 2025, so the real task is a two-month forecast. A random split would mix future days into the training set and give an over-optimistic score: in my test a random 5-fold split reported 1.61% MAPE versus 1.93% for the honest time-based evaluation. I therefore used an expanding-window time split:

Fold	Train on	Test on	Purpose
1	Jan – Apr	May – Jun	Early stability check
2	Jan – Jun	Jul – Aug	Mid-year check
3 (hold-out)	Jan – Aug	Sep – Oct	Two-month look-ahead, the same gap as Nov–Dec

Hyper-parameters were chosen on these folds only (six configurations of tree depth and learning rate; the differences between them were small, about $0.5 of MAE). The final model was then refit on all of Jan–Oct. Every fold is scored two ways: “raw” on all test rows (the honest number, including corrupted labels that no model can predict) and “clean” on test rows without flagged labels.

3. Data-quality issues and how they were handled

Issue	Train / Validation	Treatment
Negative weights (sign errors)	292 / 145 rows	Converted to absolute value
Missing weight	300 / 165 rows	Median by equipment type, plus a missing-flag
Weight at the 47,500 lb cap	1,191 / 297 rows	Kept (a real ceiling), flagged
Missing market_index	374 / 249 rows	Median of the same date (market-wide daily signal)
Distance noise on the same route; floor at 70 miles	48 / 21 rows at the floor	Route-level median distance used as a feature
Corrupted labels (rate roughly 3.4x too high or 0.28x too low)	677 train rows (1.4%)	Detected with a robust regression (residual above 6 MAD); removed from training only

Corrupted labels are removed only from training data. Validation rows are never dropped because every one of them needs a prediction.

4. Model and reasoning

Exploration showed that, after accounting for the route and equipment, the log of the rate moves linearly with the daily market index and the quote signal, rises steadily over time (roughly 0.7% per month) and has small weekday effects. Tree models cannot extrapolate beyond the values they saw in training, so the market and trend effects are modelled linearly while the non-linear route, distance and equipment structure is left to boosted trees:

log(rate) = f_static(route geometry, equipment, weight) + g_time(daily market index, quote signal, trend, weekday)

- **f_static:** HistGradientBoosting (31 leaves, learning rate 0.03, 800 iterations).
- **g_time:** ridge regression; the two parts are fitted jointly by back-fitting.
- **Calendar month is not used** because November and December never appear in training; the trend and market signals carry that information instead.

5. Validation results

Mean over the three time folds (dollars; MAPE in %):

Model	Raw MAE	Raw MAPE	Clean MAE	Clean MAPE
Chosen: tree model + linear market/trend	**98.76**	**4.28**	**44.83**	**1.93**
HistGB, all features	104.86	4.49	51.01	2.14
Ridge, all features	105.13	4.58	51.28	2.23
Random forest, all features	120.62	5.14	66.91	2.82
HistGB, no market/time information	134.98	5.63	81.46	3.34
Baseline: median $/mile by equipment and distance band	158.70	6.90	105.47	4.61

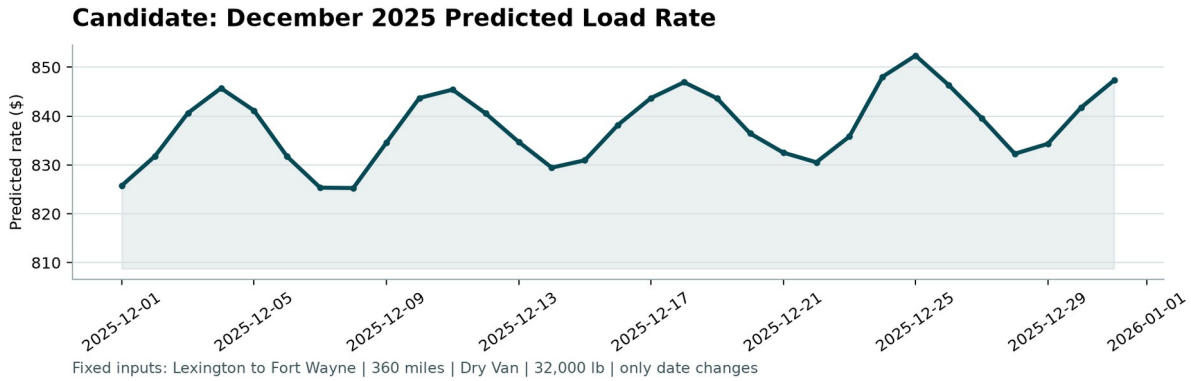
Clean MAPE by fold (stability):

Model	Fold 1	Fold 2	Fold 3 (hold-out)
Chosen: tree model + linear market/trend	**1.86**	**1.78**	**2.14**
HistGB, all features	1.95	2.50	1.97
HistGB, no market/time information	5.69	2.34	1.98
Ridge, all features	2.18	2.08	2.44

Interpretation. The chosen model has the best average error and is the most consistent across folds. On the final hold-out fold alone, the tree-only models are slightly better (about 1.97–1.98% versus 2.14%). The hold-out months had a market level below anything seen earlier in the year, which favours models that ignore the market signal. For Nov–Dec the market index lies inside the range seen in training, so I preferred the model that uses the market information and is stable across folds. The raw RMSE (about $633) is almost identical for every model because it is dominated by the corrupted labels, which cannot be predicted.

6. Fixed December 2025 prediction chart

The chart below was produced by the provided score.py for the fixed load (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb) with only the date changing. December rows contain no market_index or quote_signal, so the day-level values observed for each December date in validation.csv (features only, no labels) were used.



Predicted rates range from $825.27 (8 December) to $852.39 (25 December), with a mean of $837.95. The weekly pattern comes from the weekday effects and the daily market signal, and the level is consistent with the roughly $2.3 per mile seen in the training data for a 360-mile Dry Van load.

7. Limitations

- The upward trend is extrapolated linearly into November and December; a market change that is not visible in the market index would not be captured.
- December predictions rely on the validation file's daily market signals, because the December input file does not contain them.
- Final scoring is done by Spotter on the hidden validation labels, so the exact final metric cannot be confirmed here.

Appendix: Exploratory data overview

Distribution of posted_rate, log rate per mile (both tails show the corrupted labels), rate per mile versus distance, and the market index over time for train and validation.

